Side by Side

Use any AI tool to build two items placed side by side: two solutions, cases, concepts or sources. Students compare them, write the rule they found and try it on a new case, in about fifteen minutes.

Four steps

Compare
Fill in the three columns: only in A, in both, only in B.

Write the rule
In one or two sentences, say what the comparison shows.

Try it on case C
Use your rule on the new case and explain your answer.

Think back
Finish the sentence about how you thought, not what you found.

Two levels

Easier to see
A and B differ in one difference students can spot at once. The work is to explain why that difference matters.

Harder to see
The surface and the principle do not line up: A and B look almost the same but rest on different principles, or look different but share one principle.

Copy only the FOR STUDENTS block for students. Before you print, read the TEACHER KEY and check everything listed under CHECK. Let students write the rule before you confirm it.

How strong is the evidence?

Strong Meta-analyses or several classroom experiments show that learning by this kind of comparison works.

Promising Comparison works in related research, but few school studies test this exact kind of pair.

Strong **Learning by comparing cases.** Comparing cases side by side improves learning more than other ways of studying the same material, such as studying cases one at a time (Alfieri et al. 2013, a meta-analysis of 57 experiments, d ≈ 0.50). Learning was greater when the principle was given after the comparison, which is why students write the rule before the teacher confirms it.

Strong **Comparing solution methods.** In randomized classroom studies with Grade 7 and 8 algebra students, comparing two solution methods side by side built more flexible problem solving than studying the methods one at a time (Rittle-Johnson & Star 2007) or comparing problems solved the same way (2009). Gains in conceptual knowledge appeared in the second study but not the first.

Promising **Comparing correct and incorrect solutions.** Comparing a correct and an incorrect example helped primary students learn decimals and reduced misconceptions more than comparing two correct examples (Durkin & Rittle-Johnson 2012), and incorrect examples supported conceptual understanding in algebra classes (Booth et al. 2013). Results have not replicated consistently.

Promising **Contrasting cases with one change.** Analysing contrasting cases before a lecture helped college students notice what matters and learn more from the explanation that followed (Schwartz & Bransford 1998).

Promising **Concepts that are easily confused.** Mixing examples from different categories (interleaving) helps learners tell them apart (Brunmair & Richter 2019, a meta-analysis of 59 studies, g = 0.42), especially when the categories are similar. Few studies have tested confusable concept pairs side by side in school.

Promising **Comparing sources.** In a six-month quasi-experiment in five urban high schools, Grade 11 students taught with document-based lessons, which involved reading and questioning several sources on the same event, outperformed control classes in historical thinking, factual knowledge and reading comprehension (Reisman 2012). The study tested the whole curriculum, not source comparison alone.

Labels rate the research on teaching with comparisons regularly, not a single use. One Side by Side is one small piece of that practice.

PAIR 1

Two ways to the same answer

Two correct methods reach the same answer by different routes.

WHAT TO LOOK FOR

- Where do the two methods first part ways?
- What does each method save, and what does it cost?
- When would you choose each one?

Maths · Grade 8 · Easier to see

Task: Solve 3(x + 2) = 15.

A · ALEX

3x + 6 = 15
3x = 9
x = 3

B · SAM

x + 2 = 5
x = 3

ONLY A

Alex multiplies out the brackets first, then undoes the + 6 and the × 3. Three lines.

BOTH

The same answer, x = 3. Both do the same thing to both sides, so the equation stays balanced.

ONLY B

Sam divides both sides by 3 first, because 15 divides exactly by 3. Two lines.

THE RULE	Dividing first is quicker when the number outside the brackets divides the other side exactly. Otherwise, multiplying out first avoids working with fractions early on.
CASE C	Solve 3(x + 4) = 20. Which method would you use, and why?
CASE C ANSWER	Multiply out first, because 20 does not divide exactly by 3: 3x + 12 = 20, 3x = 8, x = 8/3. Dividing first gives x + 4 = 20/3, and then you have to work out 20/3 − 4.
PROBE	Change 3(x + 2) = 15 so that Sam's first step becomes the harder one. What did you change?
THINK BACK	Before today, my first step was always ___.

For the AI. Show two correct methods for the same task, each written as a fictional student's working. The methods differ in the first or main step and reach the same correct answer. Choose a case C where one method is clearly the better choice.

Evidence. Strong Comparing cases in general: Strong. Labels rate research on teaching with comparisons regularly, not a single use.

PAIR 2

Right next to wrong

A correct and an incorrect solution to the same task, side by side.

WHAT TO LOOK FOR

- Find the first line where the two solutions differ.
- Say which rule the wrong step breaks.
- Check that the correct solution keeps that rule.

Chemistry · Grade 9 · Easier to see

Task: Balance the equation for hydrogen burning in oxygen to make water: $H_2 + O_2 \rightarrow H_2O$.

A · MIA

$2H_2 + O_2 \rightarrow 2H_2O$
Left: 4 H, 2 O. Right: 4 H, 2 O.

B · LEO

This solution has an error.
$H_2 + O_2 \rightarrow H_2O_2$
Left: 2 H, 2 O. Right: 2 H, 2 O.

ONLY A

Mia changes the coefficients, the numbers in front of the formulas, and keeps the product as H_2O.

BOTH

Both start from the same reactants, and both end with the same number of each atom on each side.

ONLY B

Leo changes a subscript, the small number inside the formula, so the product becomes H_2O_2, hydrogen peroxide, not water.

THE RULE	Balance an equation by changing the coefficients (the numbers in front of the formulas), never the subscripts (the small numbers inside them). Changing a subscript changes the substance.
CASE C	Balance the equation for magnesium burning in oxygen: $Mg + O_2 \rightarrow MgO$.
CASE C ANSWER	$2Mg + O_2 \rightarrow 2MgO$. Writing MgO_2 would balance the oxygen but describe a different substance.
PROBE	Leo's equation is balanced. So why is it still wrong for this task?
THINK BACK	The difference I nearly missed was ___.

For the AI. Show a correct and an incorrect solution to the same task, each written as a fictional student's working. The incorrect one rests on one common misconception, and its first line is "This solution has an error." Apart from that, keep the two as alike as possible.

Evidence. Promising Comparing cases in general: Strong. Labels rate research on teaching with comparisons regularly, not a single use.

PAIR 3

One thing changed

Two cases are identical except for one thing.

WHAT TO LOOK FOR

- Name the one thing that is different.
- Compare what happened, not just what was done.
- Ask what the difference in results tells you about that one thing.

Biology · Grade 7 · Harder to see

Task: Two dishes of cress seeds were set up on Monday. Compare them on Friday.

A · UNCOVERED DISH	B · COVERED DISH
Cress seeds on damp cotton wool, uncovered on a windowsill at room temperature. On Friday the seeds have sprouted: short seedlings with green seed leaves.	Cress seeds on damp cotton wool, on the same windowsill, covered by a light-proof box. On Friday the seeds have sprouted: tall, thin seedlings with pale yellow seed leaves.

ONLY A	BOTH	ONLY B
Light. The seedlings are short with green seed leaves.	Water, warmth, the same windowsill and the same seeds. Both dishes have sprouted.	Darkness under the box. The seedlings are tall and thin with pale yellow seed leaves.

THE RULE	The cress seeds sprouted in both dishes, so cress seeds do not need light to sprout: the seed uses its own stored food. Light matters afterwards, for the green leaves that make new food.
CASE C	Dish C is set up like the uncovered dish, on the same windowsill, but the cotton wool is dry. What will it look like on Friday?
CASE C ANSWER	The seeds will not sprout. Sprouting needs water, even with light and warmth.
PROBE	The two dishes look very different. Why is "light makes seeds sprout" still the wrong conclusion?
THINK BACK	At first I looked at ___ instead of what both dishes had in common.

For the AI. Describe two cases, such as two experiments, places or situations, that are identical except for one variable, and say what happened in each. Do not name the changed variable as the point of the task; it belongs in the teacher key.

Evidence. Promising Comparing cases in general: Strong. Labels rate research on teaching with comparisons regularly, not a single use.

PAIR 4

Two concepts that are easily confused

Two related ideas that students often mix up, shown in the same context.

WHAT TO LOOK FOR

- List what each one has that the other does not.
- Find what they share, so you know why they get mixed up.
- Make a quick test you could use next time.

English · Grade 7 · Easier to see

Task: Two sentences describe the same hot classroom.

A · SIMILE	B · METAPHOR
The classroom was like an oven.	The classroom was an oven.

ONLY A	BOTH	ONLY B
The word "like". It says the classroom resembles an oven.	Both compare the classroom to an oven to show how hot it was. Neither is meant literally.	No comparing word. It says the classroom is an oven.

THE RULE	A simile compares two different things using "like" or "as". A metaphor says one thing is another. Both are comparisons, not literal statements.
CASE C	"The test was a mountain to climb." Simile or metaphor?
CASE C ANSWER	A metaphor: there is no "like" or "as", and it says the test is a mountain.
PROBE	Which sentence is stronger, and why might a writer still choose the simile?
THINK BACK	The test I will use next time is ___.

For the AI. Show one clear example of each of two concepts students in this grade often confuse, both in the same context, and label each with its concept. Choose examples that differ in the feature that defines each concept.

Evidence. Promising Comparing cases in general: Strong. Labels rate research on teaching with comparisons regularly, not a single use.

PAIR 5

Two sources on the same event

Two sources describe the same event but differ in when, by whom or why they were written, which changes how far we can trust them.

WHAT TO LOOK FOR

- Who wrote each source, when and why?
- Which details agree, and which do not?
- Which source would you trust for which question?

History · Grade 9 · Harder to see

Task: Two accounts of the first morning of a strike at a cotton mill in 1912. Both were written for this task.

A · DIARY, 1912

Mary Dent, mill worker, private diary, 4 May 1912: "Went to the gate at six. Perhaps two hundred of us. Mr Hale, the owner, came out and said the mill would close if we did not go in. Nobody moved. Rain all afternoon." (Written for this task.)

B · MEMOIR, 1950

Mary Dent, memoir published in 1950: "I shall never forget the morning a thousand of us stood at the gates in brilliant sunshine, and the owner fled before our courage." (Written for this task.)

ONLY A

Written the same day, in a private diary. Small, plain details: the time, the rain, about two hundred people.

BOTH

The same eyewitness and the same event: a crowd at the gate and the owner facing them.

ONLY B

Written 38 years later, for readers. Bigger numbers, sunshine and a heroic story.

THE RULE	Even the same eyewitness becomes less reliable on details when writing long after the event for an audience. For details, prefer accounts written close to the event; later accounts show how the event was remembered.
CASE C	A union leaflet printed the next day, calling on other workers to join, says "thousands stood firm at Hale's gate". How far would you trust its number?
CASE C ANSWER	Not far. It was written close to the event, but to persuade, so the number may be inflated. Check it against Source A.
PROBE	What can Source B tell us that Source A cannot?
THINK BACK	I trusted ___ before I checked when and why it was written.

For the AI. Write two short sources about the same event, each with an author, a date and a purpose. Mark both "(written for this task)" unless you quote a real source word for word. The sources differ mainly in one thing that affects how far we can trust them, such as when, by whom or why they were written.

Evidence. Promising Comparing cases in general: Strong. Labels rate research on teaching with comparisons regularly, not a single use.

WITHOUT THE WEBSITE

Your own pair

Fill in the blanks, then type the prompt into any AI tool: Copilot, Gemini, ChatGPT or Claude. Check the key before the lesson, especially anything under CHECK.

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Do not include student names or student work in this conversation.

You are helping a Grade _____ _____ teacher prepare a fifteen-minute "Side by
Side" task. Students compare two items placed next to each other: what is only in A, what is in
both, and what is only in B. Then they write the rule they found and try it on a new case C.
Students work on paper, without AI.

Lesson topic: _____
Students already know (optional): _____

Pair type: _____ (copy the name and one-line description from its
card)
How to build the pair: _____ (copy the "For the AI" text from its card)
Difficulty: _____ (copy one level from the cover)

Rules: A and B respond to the same task and are as alike as possible apart from the difference this
pair type asks for. Each of A and B is at most 60 words or six short lines, so it fits half a page
width. Case C can be decided with the rule. Do not invent studies, measurements or sources; a
source written for the task must say so. Write for students in this grade.

Reply in plain text, no tables, in two blocks. The first block is printed for students, so it must
not contain anything from the second.
FOR STUDENTS
TASK: the question or task both A and B respond to
A: a short label, then the first item
B: a short label, then the second item
CASE C: a new case the rule decides, without its answer
THINK BACK: one short sentence starter with a single blank, about how they thought, not what they
found
TEACHER KEY
ONLY A: what only A has or does
BOTH: what A and B share
ONLY B: what only B has or does
THE RULE: the principle students should reach, in one or two sentences
CASE C ANSWER: the answer, and how the rule decides it
PROBE: one follow-up question that pushes past the first answer
CHECK: only the facts, figures, dates or sources in TASK, A, B or CASE C that could be wrong,
quoted or closely paraphrased, most doubtful first, one per line, at most six lines. Write "None"
if there are none.
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Faster online: the prompt builder at choosewise.education/side-by-side fills this in for you. Students need no AI for the main task. If older students use AI themselves, check the tool's age rules, your national rules and your school policy first.